

GCL BENTOMAT® DN

REINFORCED GEOSYNTHETIC CLAY LINER (GCL)

PROPERTIES	TEST METHOD	GCL BENTOMAT® – BDN
		IMPERIAL
¹ BENTONITE MOISTURE CONTENT	ASTM D2216	12% max
¹ BENTONITE SWELL INDEX	ASTM D5890	24 mL/2g min
¹ BENTONITE FLUID LOSS	ASTM D5891	18 mL max
² BENTONITE MASS/AREA	ASTM D5993	0.75 lb/ft ² (3.7 kg/m ²) min
² TOTAL MASS/AREA	ASTM D5993	0.83 lb/ft ² (4.1 kg/m ²) min
GCL MOISTURE CONTENT	ASTM D5993	35% max
³ GCL GRAB STRENGTH	ASTM D6768	50 lbs/in (8.8 kN/m) min
GCL PEEL STRENGTH	ASTM D6496	3.5 lbs/in (610 N/m) min
⁴ GCL HYDRAULIC CONDUCTIVITY	ASTM D5887	5 x 10 ⁻¹¹ m/s max
⁴ GCL INDEX FLUX	ASTM D5887	1 x 10 ⁻⁸ m ³ /m ² /s max
⁵ GCL Hydrated Internal Shear Strength	ASTM D6243	500 psf (24 kPa) typ.@ 200 psf (9.6 kPa)

Bentomat® is a registered trademark of Cetco®.

¹ Bentonite property tests performed before the bentonite is incorporated into the finished GCL product.

² Reported at 0 percent moisture content.

³ All tensile strength testing is performed in the machine direction using ASTM D6768.

⁴ Index flux and hydraulic conductivity testing with deaired distilled/deionized water at 80 psi (550 kPa) cell pressure, 77 psi (530 kPa) headwater pressure and 75 psi (515 kPa) tailwater pressure.

⁵ Peak values measured at 200 psf (9.6 kPa) normal stress for a specimen hydrated for 48 hours. Site-specific materials, GCL products, and test conditions must be used to verify internal and interface strength of the proposed design.

BENTOMAT®

Bentomat® DN is a reinforced bentonite-based geosynthetic clay liner (GCL) comprised of two polypropylene nonwoven geotextiles that are needle-punched together, encapsulating a layer of sodium bentonite clay between the two textiles. GCLs are a great secondary or supportive layer used in conjunction with the right geomembrane. GCLs are known to provide consistent, very-low permeability and have the unique ability to seal around penetrations, self-heal punctures, and self-seam at the overlaps. Bentonite needle-punched GCLs provide a durable reinforcement to withstand shear stresses on steep slopes and the high needle-punch density provides higher peak internal shear strengths.

Bentomat® DN features granular bentonite, which creates less dust during installation than powdered bentonite, and is less likely to shift through the needle-punched reinforcement providing a consistent hydraulic performance. Standard bentonite-based GCLs are designed to provide a hydraulic barrier against many leachates. The swelling bentonite fills up pore space and constricts flow paths to water, resulting in a low-permeability hydraulic barrier. Bentomat® DN is frequently used in environmental containment applications and is a high-performance, cost-effective alternative to compacted clay liners.

Note: To the best of our knowledge, unless otherwise stated, these are typical property values and are intended as guides only, not as specification limits. Chemical resistance, odor transmission, longevity as well as other performance criteria is not implied or given and actual testing must be performed for applicability in specific applications and/or conditions. RAVEN INDUSTRIES MAKES NO WARRANTIES AS TO THE FITNESS FOR A SPECIFIC USE OR MERCHANTABILITY OF PRODUCTS REFERRED TO, no guarantee of satisfactory results from reliance upon contained information or recommendations and disclaims all liability for resulting loss or damage.

RAVEN ENGINEERED FILMS

P.O. Box 5107 Sioux Falls, SD 57117-5107
Ph: +1 (605) 335-0174 • TF: +1 (800) 635-3456

© 2018 RAVEN INDUSTRIES INC. All rights reserved.

efdsales@ravenind.com
www.ravenefd.com

RAVEN

060618 EFD 3132